

Cognitive Robotics Lab 2

Piyush R Saini (23BRS1150)

Videos: <https://github.com/findpiyush/cognitive-robotics>

Using a python script to move the robot

Code:

```
#!/usr/bin/env python3
```

```
import sys
```

```
import select
```

```
import termios
```

```
import tty
```

```
import rclpy
```

```
from rclpy.node import Node
```

```
from geometry_msgs.msg import Twist
```

```
class KeyboardTeleop(Node):
```

```
    def __init__(self):
```

```
        super().__init__('keyboard_teleop')
```

```
        self.publisher = self.create_publisher(
```

```
            Twist,
```

```

        '/cmd_vel',
        10
    )

    # Speeds
    self.linear_speed = 0.20
    self.angular_speed = 0.80

    # Publish continuously
    self.timer = self.create_timer(
        0.1,
        self.publish_command
    )

    self.current_command = Twist()

    self.get_logger().info('Keyboard teleop started')

def publish_command(self):
    self.publisher.publish(self.current_command)

def set_command(self, linear, angular):
    self.current_command.linear.x = linear
    self.current_command.linear.y = 0.0
    self.current_command.linear.z = 0.0

    self.current_command.angular.x = 0.0

```

```

        self.current_command.angular.y = 0.0
        self.current_command.angular.z = angular

    def stop(self):
        self.set_command(0.0, 0.0)

def get_key(settings):
    tty.setraw(sys.stdin.fileno())

    key = sys.stdin.read(1)

    termios.tcsetattr(
        sys.stdin,
        termios.TCSADRAIN,
        settings
    )

    return key

def main(args=None):

    rclpy.init(args=args)

    node = KeyboardTeleop()

    settings = termios.tcgetattr(sys.stdin)

```

```

print('')
print('=====')
print('          BeetleBot Keyboard Teleop')
print('=====')
print('')
print('          w')
print('          ↑')
print('    a ← s → d')
print('          ↓')
print('          x')
print('')
print('w : Forward')
print('x : Backward')
print('a : Turn left')
print('d : Turn right')
print('s : STOP')
print('q : Quit')
print('')
print('Speed:', node.linear_speed)
print('=====')
print('')

```

```

try:

```

```

    while rclpy.ok():

```

```

        key = get_key(settings)

```

```
if key == 'w':
    node.set_command(
        node.linear_speed,
        0.0
    )

elif key == 'x':
    node.set_command(
        -node.linear_speed,
        0.0
    )

elif key == 'a':
    node.set_command(
        0.0,
        node.angular_speed
    )

elif key == 'd':
    node.set_command(
        0.0,
        -node.angular_speed
    )

elif key == 's':
    node.stop()
```

```

        elif key == 'q':
            node.stop()
            break

        # Let ROS process timers
        rclpy.spin_once(
            node,
            timeout_sec=0.01
        )

except KeyboardInterrupt:
    pass

finally:

    # Always stop the robot
    node.stop()

    # Publish several stop messages
    for _ in range(5):
        rclpy.spin_once(
            node,
            timeout_sec=0.02
        )

    termios.tcsetattr(

```

```

        sys.stdin,
        termios.TCSADRAIN,
        settings
    )

    node.destroy_node()

    rclpy.shutdown()

if __name__ == '__main__':
    main()

```

Output Screenshots:

```

student@jazzy-jalisco:~$ cd beetlebot_ws/
student@jazzy-jalisco:~/beetlebot_ws$ cd src/
student@jazzy-jalisco:~/beetlebot_ws/src$ ros2 pkg create --build-type ament_python keyboard_teleop --dependencies rclpy geometry_msgs
going to create a new package
package name: keyboard_teleop
destination directory: /home/student/beetlebot_ws/src
package format: 3
version: 0.0.0
description: TODO: Package description
maintainer: ['student <student@todo.todo>']
licenses: ['TODO: License declaration']
build type: ament_python
dependencies: ['rclpy', 'geometry_msgs']
creating folder ./keyboard_teleop
creating ./keyboard_teleop/package.xml
creating source folder
creating folder ./keyboard_teleop/keyboard_teleop
creating ./keyboard_teleop/setup.py
creating ./keyboard_teleop/setup.cfg
creating folder ./keyboard_teleop/resource
creating ./keyboard_teleop/resource/keyboard_teleop
creating ./keyboard_teleop/keyboard_teleop/__init__.py
creating folder ./keyboard_teleop/test
creating ./keyboard_teleop/test/test_copyright.py
creating ./keyboard_teleop/test/test_flake8.py
creating ./keyboard_teleop/test/test_pep257.py

[WARNING]: Unknown license 'TODO: License declaration'. This has been set in the package.xml, but no LICENSE file has been created.
It is recommended to use one of the ament license identifiers:
Apache-2.0
BSL-1.0
BSD-2.0
BSD-2-Clause
BSD-3-Clause
GPL-3.0-only
LGPL-3.0-only
MIT
MIT-0
student@jazzy-jalisco:~/beetlebot_ws/src$ cd ~/beetlebot_ws/src/keyboard_teleop
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$ nano keyboard_teleop/teleop.py
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$ chmod +x ~/beetlebot_ws/src/keyboard_teleop/keyboard_teleop/teleop.py
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$ cd ~/beetlebot_ws/src/keyboard_teleop
nano setup.py

```

```
INFO
student@jazzy-jalisco:~/beetlebot_ws/src$ cd ~/beetlebot_ws/src/keyboard_teleop
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$ nano keyboard_teleop/teleop.py
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$ chmod +x ~/beetlebot_ws/src/keyboard_teleop/keyboard_teleop/teleop.py
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$ cd ~/beetlebot_ws/src/keyboard_teleop
nano setup.py
student@jazzy-jalisco:~/beetlebot_ws/src/keyboard_teleop$ cd ..
student@jazzy-jalisco:~/beetlebot_ws/src$ cd ..
student@jazzy-jalisco:~/beetlebot_ws$ colcon build --packages-select keyboard_teleop
Starting >>> keyboard_teleop
Finished <<< keyboard_teleop [0.81s]

Summary: 1 package finished [0.90s]
student@jazzy-jalisco:~/beetlebot_ws$ source ~/beetlebot_ws/install/setup.bash
student@jazzy-jalisco:~/beetlebot_ws$ ros2 pkg list | grep keyboard_teleop
keyboard_teleop
student@jazzy-jalisco:~/beetlebot_ws$ ros2 pkg executables keyboard_teleop
keyboard_teleop teleop
student@jazzy-jalisco:~/beetlebot_ws$
```